

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException AtTheRateException DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

```
// You are using Java
import java.util.Scanner;
```

```
class DotException extends Exception {
    public DotException(String message) {
        super(message);
    }
}
```

```
class AtTheRateException extends Exception {
    public AtTheRateException(String message) {
        super(message);
    }
}
```

```
class DomainException extends Exception {
    public DomainException(String message) {
        super(message);
    }
}
```

```
public class Main {
    public static void validateEmail(String email) throws DotException,
    AtTheRateException, DomainException {
        // Check for exactly one '@'
        int atCount = 0;
        for (char ch : email.toCharArray()) {
            if (ch == '@') atCount++;
        }
        if (atCount != 1) throw new AtTheRateException("AtTheRateException:
        Invalid @ usage");
    }
}
```

```

// Check email doesn't start or end with '.' or '@'
if (email.startsWith(".") || email.startsWith("@") || email.endsWith(".") ||
email.endsWith("@")) {
    throw new DotException("DotException: Invalid Dot usage");
}

// Check no consecutive '.' or '@'
for (int i = 1; i < email.length(); i++) {
    if ((email.charAt(i) == '.' && email.charAt(i-1) == '.') ||
        (email.charAt(i) == '@' && email.charAt(i-1) == '@')) {
        if (email.charAt(i) == '.') throw new DotException("DotException: Invalid
Dot usage");
        else throw new AtTheRateException("AtTheRateException: Invalid @
usage");
    }
}

int atIndex = email.indexOf('@');
String domainPart = email.substring(atIndex + 1);

// There must be at least one '.' after '@'
if (!domainPart.contains(".")) {
    throw new DotException("DotException: Invalid Dot usage");
}

// The domain extension should be after the last '.'
int lastDotIndex = domainPart.lastIndexOf('.');
if (lastDotIndex == domainPart.length() - 1) {
    throw new DotException("DotException: Invalid Dot usage");
}
String domainExtension = domainPart.substring(lastDotIndex + 1);

if (!(domainExtension.equals("in") || domainExtension.equals("com") ||
domainExtension.equals("net") || domainExtension.equals("biz"))) {
    throw new DomainException("DomainException: Invalid Domain");
}

}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    String email = sc.nextLine();

```

```
try {
    validateEmail(email);
    System.out.println("Valid email address");
} catch (DotException e) {
    System.out.println(e.getMessage());
    System.out.println("Invalid email address");
} catch (AtTheRateException e) {
    System.out.println(e.getMessage());
    System.out.println("Invalid email address");
} catch (DomainException e) {
    System.out.println(e.getMessage());
    System.out.println("Invalid email address");
}
}
sc.close();
}
```

Status : Correct

Marks : 10/10